

Time-Resolved Asymmetry: Human-AI vs Human-Human vs AI-AI

How story-internal asymmetry evolves across early / middle / late phases

PENPAL Project

2026-05-06

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1 Overview

This notebook tracks how within-dyad asymmetry evolves *across the story*. For each metric (valence, novelty-resonance slope, semantic-exploration AUC), we compute the per-story signed and unsigned slot-1 vs slot-2 difference within rolling thirds (Early / Middle / Late) of the story turn window, and test whether condition $\times$ phase interactions appear.

Two outcomes per metric:

- **Magnitude:** per-story $|A|$ in each phase.
- **Direction:** per-story signed slot-difference in each phase (with the identity-tied directionality reading: HA should keep a stable sign across phases; HH and AA should show no fixed sign).

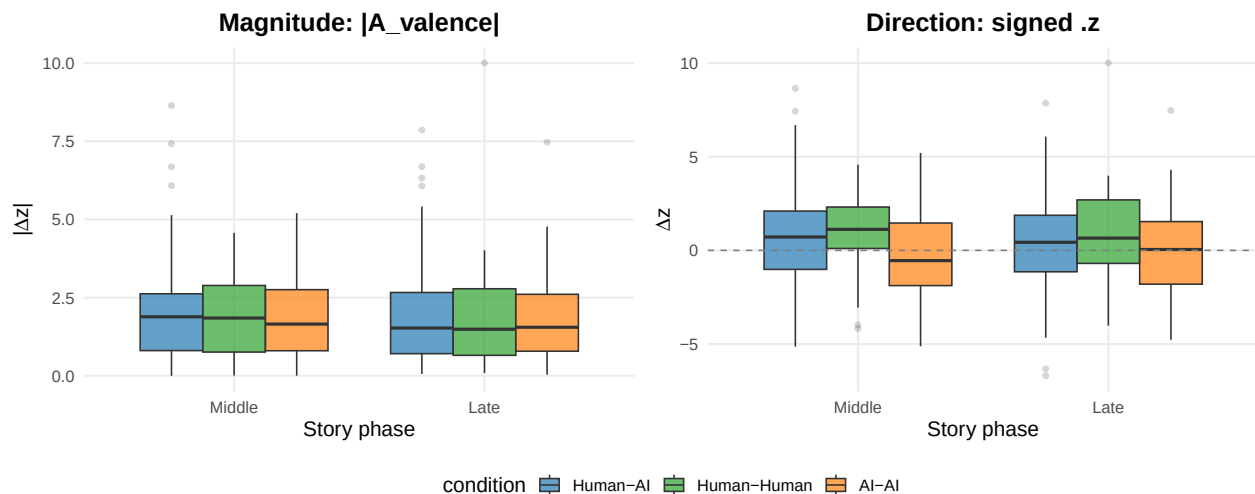
2 Phase definition

3 Valence Asymmetry by Phase

Table 1: Per-story valence asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	94	0.605	2.602	2.079
Human-AI	Late	98	0.340	2.518	1.938
Human-Human	Middle	36	0.859	2.247	1.982
Human-Human	Late	36	0.833	2.575	1.967
AI-AI	Middle	80	-0.348	2.269	1.875
AI-AI	Late	80	-0.003	2.309	1.835

condition	phase	n	mean_signed	sd_signed	mean_abs
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```
## LMM: A_valence ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_valence ~ condition * phase + (1 | conversation_id)
## Data: valence_phase_A
##
## REML criterion at convergence: 1568.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.3336 -0.7921 -0.1699  0.5113  5.1832
##
## Random effects:
##  Groups             Name             Variance Std.Dev.
##  conversation_id (Intercept) 0.04718  0.2172
##  Residual                2.30383  1.5178
## Number of obs: 424, groups: conversation_id, 215
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      2.07982    0.15815 417.86398   13.151  <2e-16
## conditionHuman-Human -0.09744    0.30053 417.84063   -0.324    0.746
## conditionAI-AI      -0.20452    0.23323 417.84671   -0.877    0.381
## phaseLate         -0.14174    0.21920 212.22232   -0.647    0.519
## conditionHuman-Human:phaseLate 0.12665    0.41957 207.38047    0.302    0.763
## conditionAI-AI:phaseLate  0.10142    0.32503 208.58119    0.312    0.755
##
## (Intercept)                ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate
## conditionAI-AI:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

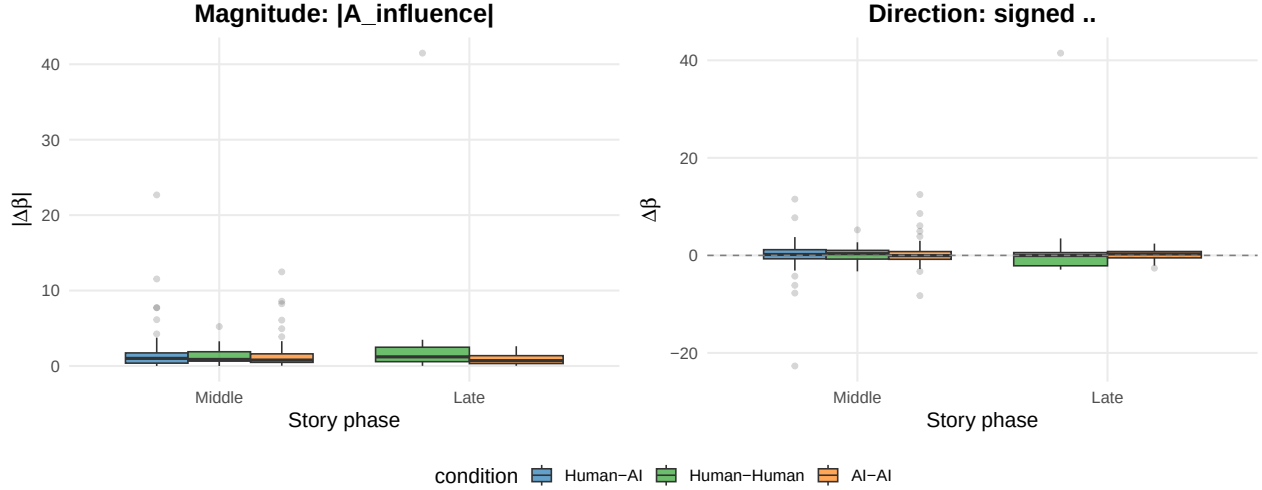
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.526
## condtnAI-AI -0.678  0.357
## phaseLate   -0.708  0.372  0.480
## cndtnHm-H:L  0.370 -0.702 -0.251 -0.522
## cndtAI-AI:L  0.477 -0.251 -0.704 -0.674  0.352
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
## Data: valence_phase_A
##
## REML criterion at convergence: 1955.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.8858 -0.6203  0.0188  0.6454  3.7655
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept)  0.000         0.000
## Residual                    5.937         2.437
## Number of obs: 424, groups: conversation_id, 215
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.6053    0.2513 418.0000   2.409  0.0164 *
## conditionHuman-Human  0.2542    0.4776 418.0000   0.532  0.5948
## conditionAI-AI      -0.9536    0.3706 418.0000  -2.573  0.0104 *
## phaseLate        -0.2653    0.3518 418.0000  -0.754  0.4512
## conditionHuman-Human:phaseLate  0.2386    0.6735 418.0000   0.354  0.7233
## conditionAI-AI:phaseLate  0.6101    0.5217 418.0000   1.170  0.2429
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.526
## condtnAI-AI -0.678  0.357
## phaseLate   -0.714  0.376  0.484
## cndtnHm-H:L  0.373 -0.709 -0.253 -0.522
## cndtAI-AI:L  0.482 -0.254 -0.710 -0.674  0.352
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

4 Novelty-Resonance Slope Asymmetry by Phase

Table 2: Per-story novelty-resonance slope asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	95	0.124	3.299	1.711
Human-Human	Middle	36	0.260	1.685	1.291
Human-Human	Late	17	2.028	10.305	3.729
AI-AI	Middle	80	0.224	2.459	1.429
AI-AI	Late	39	0.173	1.180	0.939



```
## LMM: A_influence ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_influence ~ condition * phase + (1 | conversation_id)
## Data: novelty_phase_A
##
## REML criterion at convergence: 1369.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.1628 -0.3247 -0.1890  0.0642 11.8478
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept)  0.00    0.000
## Residual                10.15    3.186
## Number of obs: 267, groups: conversation_id, 211
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      1.7109    0.3269 262.0000   5.234 3.4e-07
## conditionHuman-Human -0.4198    0.6235 262.0000  -0.673 0.50138
## conditionAI-AI      -0.2822    0.4834 262.0000  -0.584 0.55996
## phaseLate         -0.4899    0.6222 262.0000  -0.787 0.43172
## conditionHuman-Human:phaseLate  2.9283    1.1252 262.0000   2.602 0.00978
##
```

```

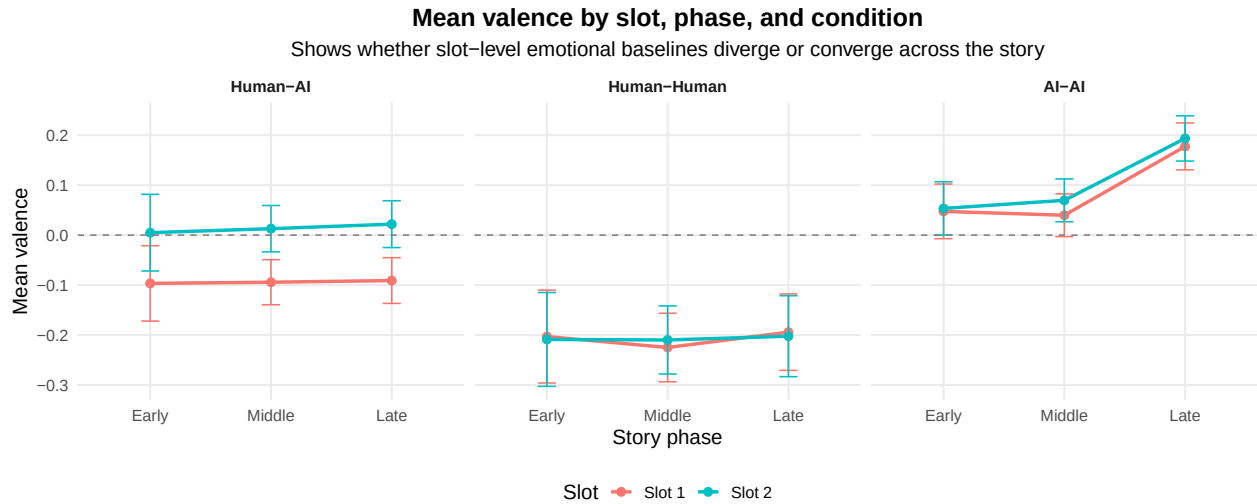
## (Intercept) ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.524
## condtnAI-AI -0.676  0.354
## phaseLate   0.000  0.000 -0.422
## cndtnHm-H:L 0.000 -0.402  0.233 -0.553
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
##   Data: novelty_phase_A
##
## REML criterion at convergence: 1430.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.3742 -0.3071 -0.0321  0.2178 11.0276
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept)  0.0         0.000
## Residual                  12.8         3.577
## Number of obs: 267, groups: conversation_id, 211
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)    0.12397    0.36700 262.00000   0.338   0.736
## conditionHuman-Human  0.13637    0.70009 262.00000   0.195   0.846
## conditionAI-AI      0.09965    0.54280 262.00000   0.184   0.854
## phaseLate      -0.05018    0.69859 262.00000  -0.072   0.943
## conditionHuman-Human:phaseLate  1.81767    1.26339 262.00000   1.439   0.151
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.524
## condtnAI-AI -0.676  0.354
## phaseLate   0.000  0.000 -0.422
## cndtnHm-H:L 0.000 -0.402  0.233 -0.553
## fit warnings:

```

```
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

5 Mean Valence Trajectory by Slot and Condition

A descriptive companion view: how does each slot’s mean valence drift across phase, by condition? Useful for the “story-evolution” narrative without relying on the asymmetry abstraction.



6 Summary

What this notebook adds beyond the cross-section comparisons:

- **Per-phase magnitude $|A|$** for valence and novelty-resonance: tests whether asymmetry strengthens, weakens, or stays flat across the story.
- **Per-phase signed asymmetry:** combined with per-story directional-stability, this is the strongest visual evidence for HA-specific identity-tied directionality. A signed-asymmetry boxplot that stays *consistently above zero* in HA across all three phases, while HH/AA hover around zero, is the publication-quality figure for the reframed RQ1.
- **Mean valence trajectory by slot and condition:** a story-evolution view that does not depend on the asymmetry abstraction and helps illustrate qualitative drift.

Open additions worth considering:

- Add a per-phase semantic-exploration AUC view (the lag k axis is itself an evolution axis, but a phase split could surface fine-grained shifts).
- Replace fixed thirds with overlapping rolling windows for smoother trajectories.

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